CIS Business Economics - Week 2: Twin Deficits Under

TWIN DEFICITS UNDER FIRE

The Middle East Energy Shock and India's Fiscal and Current Account Balances

CIS Business Economics (BE-2) - Week 2 Research Submission
Prepared by: M PRAVEEN KUMAR, PGPM 2025 – 26
Course: Business Economics (BE-2)
Submission Type: Independent Case Study

Core Question

"To what extent can a sustained Middle East energy shock widen India's twin deficits, and what policy mix can contain the resulting external and fiscal pressures?"

Abbreviations

Abbreviation	Full Term
CAD	Current Account Deficit
CPI	Consumer Price Index
DBT	Direct Benefit Transfer
DEA	Department of Economic Affairs
ECB	External Commercial Borrowing
FCNR(B)	Foreign Currency Non-Resident (Bank)
FDI	Foreign Direct Investment
FPI	Foreign Portfolio Investor
FRBM	Fiscal Responsibility and Budget Management (Act)
FY	Fiscal Year
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
MEA	Ministry of External Affairs
MOSPI	Ministry of Statistics and Programme Implementation
MPC	Monetary Policy Committee
MoF	Ministry of Finance
MoPNG	Ministry of Petroleum & Natural Gas
NRI	Non-Resident Indian
OMC	Oil Marketing Company
PMUY	Pradhan Mantri Ujjwala Yojana
PPAC	Petroleum Planning & Analysis Cell
RBI	Reserve Bank of India
SEBI	Securities and Exchange Board of India
SPR	Strategic Petroleum Reserve
WPI	Wholesale Price Index

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1. Executive Summary

The escalation of conflict across the Middle East - spanning Gulf tensions, Red Sea shipping attacks, and the risk of a **Strait of Hormuz** disruption - reopens a familiar problem for Indian policymakers: energy-price-driven pressure on both the **Current Account Deficit** (CAD) and the **Fiscal Deficit** simultaneously. India imports **87-88% of its crude oil**, much of it from or transiting through Middle Eastern chokepoints.

This report models three escalation scenarios based on **Brent crude** price shocks of **+$10**, **+$20** and **+$30** per barrel, using an FY25 fiscal deficit baseline of **4.90% of GDP** and a nominal GDP of approximately Rs 326.4 lakh crore (~USD 3.9 trillion).

Key Findings

Channel	Scenario A (+$10/bbl)	Scenario C (+$30/bbl, severe)
Additional oil import bill	~USD 17 billion	~USD 51 billion
Current Account Deficit	1.18% - > 1.62% of GDP (+0.44pp)	1.18% - > 2.49% of GDP (+1.31pp)
Additional subsidy burden	Rs 33,000 crore	Rs 94,000 crore
Fiscal deficit	4.90% - > 5.00% of GDP (+0.10pp)	4.90% - > 5.19% of GDP (+0.29pp)
Import cover	9.8 - > 9.6 months	9.8 - > 9.2 months

India retains genuine protection: approximately 10 months of **import cover**, a large and diversifying economy, and direct operational experience managing a comparable shock in 2022. This cushion buys time and reduces crisis probability, but it does not remove the underlying exposure - remittance and shipping-cost channels (Section 6) add pressure that a simple oil-price test misses, and the two deficits widen together not because one causes the other, but because both are driven by the same **external shock** (Section 4). Coordinated, targeted RBI and Ministry of Finance action - rather than reactive, broad-based measures - is the central recommendation of this report (Sections 12-13).

2. CIS Question, Objectives, Scope & Methodology

CIS Question: How does the Middle East energy shock strain India's Balance of Payments and simultaneously widen the fiscal deficit through subsidy and support costs, and what does this combined 'twin deficit' pressure mean for policy?

Objectives: (1) quantify the strain on India's **Balance of Payments** from the oil, shipping and remittance channels (Section 6); (2) estimate the associated **Fiscal Deficit** burden using a historically anchored subsidy-escalation model (Section 7); (3) establish - and correctly bound - the causal relationship between the two deficits rather than relying on the accounting identity alone (Section 4); (4) map the institutions that administer each channel (Section 5); and (5) convert the analysis into time-bound, trigger-based policy recommendations (Section 13). Methodology: baseline establishment (Section 3), transmission

mapping (Sections 4-5), quantitative modelling (Sections 6-7), scenario and sensitivity testing (Sections 10-11), and policy synthesis (Sections 12-13) - each step documented with its evidence type in Section 16.

This report quantifies both channels using the same **+$10/+$20/+$30** per barrel scenario ladder used in the Week 1 report, so that the two reports' findings are directly comparable. It then examines how the channels interact - not by treating the twin-deficit accounting identity as a causal proof (a distinction developed explicitly in Section 4), but by tracing the common shock that drives both deficits simultaneously. The capital-flow and reserve-adequacy consequences of this pressure are developed further in the Week 3 report; the resulting monetary and fiscal policy choice is resolved in the Week 4 Policy Defence Memo.

2.1 Confidence and Limitations

This report combines observed institutional data (Section 16.A) with scenario-calibrated estimates (Section 16.D), the two kept explicitly distinct. Confidence is higher in the direction of each effect (both deficits widen) than in its exact magnitude, since the fiscal estimates are calibrated against the 2022 energy-shock precedent rather than derived from a live, real-time data feed. Readers should treat the specific rupee figures in this report as defensible planning estimates, not precise forecasts.

3. India's Fiscal and External Baseline

These figures establish the starting position against which all scenario impacts in this report are measured. Full source detail is in Section 16.A.

Indicator	Value	Basis
Nominal GDP (FY25, BE)	Rs 326.4 lakh crore (~USD 3.90 trillion)	Union Budget 2024-25
Fiscal deficit target (FY25, BE)	4.90% of GDP	Union Budget 2024-25
Fiscal glide-path target (FY26)	4.50% of GDP	Budget Speech
Current Account Deficit (FY24)	0.7% of GDP (~USD 23 billion)	RBI
Current Account Deficit (FY25 estimate)	1.15-1.20% of GDP (~USD 46 billion)	Author estimate, RBI-consistent
Crude oil import dependency	87-88%	Ministry of Petroleum
Net crude imports (FY24)	232.5 million tonnes	PPAC
Forex reserves	~USD 655 billion	RBI
Import cover	~9.8 months	Author calculation
USD/INR reference rate	~Rs. 84/USD	Market reference, FY25
Fertiliser subsidy (FY25, BE)	~Rs 1,64,000 crore	Union Budget
Total inward remittances	~USD 125 billion	World Bank / RBI

Indicator	Value	Basis
Gulf-origin remittance share	~50% (~USD 63 billion)	Author estimate

Data Vintage & Baseline Reconciliation Note

This report's USD/INR reference (~Rs. 84/USD) and reserve figure (~**USD 655 billion**) reflect an FY25 reference period, consistent with the fiscal-year framing of this report's baseline (Table above). The Week 1 report's USD/INR baseline (~Rs. 86.0/USD) and Week 3's assignment-specified baseline (~Rs. 87.50/USD, USD 716 billion reserves) belong to later points on the same escalation timeline rather than representing a contradiction; each report is internally consistent with the reference period stated in its own baseline section. This report uses the FY25 Budget Estimate (**4.90% of GDP**) for the fiscal deficit rather than Week 1's FY26 target (4.4%) because the Week 2 CIS deliverable specifically requires estimating strain against 'recent Ministry of Finance or policy reports,' for which the FY25 BE is the applicable reference document at the time the subsidy history (Section 8) was compiled.

4. Twin-Deficit Economic Framework

This report treats the oil shock as affecting India's external and fiscal balances through four channels: (1) the direct oil import bill, which widens the **Current Account Deficit**; (2) higher fertiliser and LPG subsidy costs, which widen the **Fiscal Deficit**; (3) Red Sea shipping disruption, which raises non-oil trade costs independent of the oil price; and (4) Gulf labour-market and remittance stress, a channel unique to a Middle East-centred shock rather than a generic oil-price rise.

This report's framing connects to the long-standing 'twin-deficit hypothesis' debate in open-economy macroeconomics: the Mundell-Fleming small-open-economy view predicts that a widening fiscal deficit systematically pulls the current account wider through higher interest rates and import demand, while Ricardian-equivalence-style arguments - and a mixed empirical record for economies including India - suggest the two balances can move largely independently. This report does not need to adjudicate that debate. The claim developed below is narrower and more directly testable: a single **common external shock** (the Middle East energy shock) independently widens both balances through two separately traceable channels, a claim that holds regardless of one's position on the general twin-deficit hypothesis.

4.1 The Twin-Deficit Identity - What It Proves and What It Does Not

A standard macroeconomic identity links the Current Account and the government's fiscal position: CAD = Fiscal Deficit + (Private Investment − Private Saving). This is an accounting identity - it holds by construction, after the fact, given how the national accounts are defined. It is important to be precise about what this identity does and does not establish.

What the identity tells us: If private investment and private saving are held roughly constant, an increase in the fiscal deficit is arithmetically associated with an increase in the Current Account Deficit. This is a statement about how three accounting balances must sum to zero-ish (given valuation and statistical-discrepancy caveats), not a statement about which variable moves first or why.

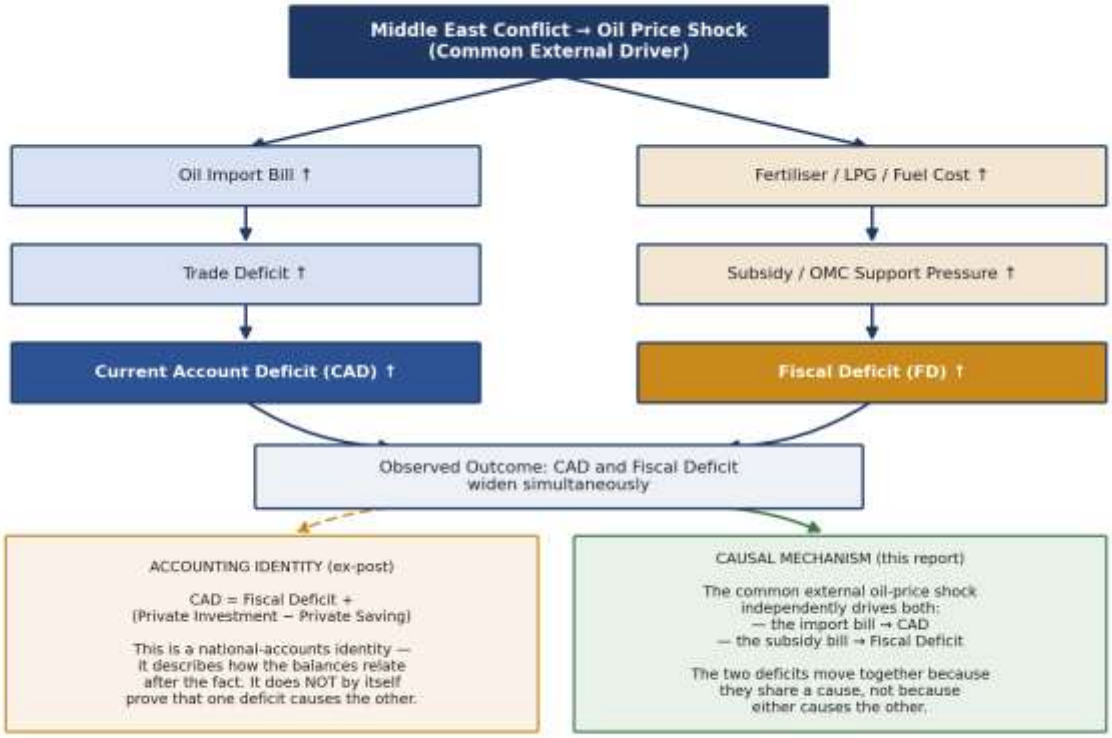
What the identity does NOT prove: The identity does not establish that the fiscal deficit causes the CAD to widen, or vice versa. Both could move together for reasons entirely unrelated to each other, or - as this report argues - because a third, common factor drives both independently. Treating the identity itself as evidence of a causal channel between the two deficits would be a basic error of economic reasoning: **an ex-post accounting relationship is not a causal mechanism**.

The actual causal mechanism in this scenario: The Middle East energy shock is the common external driver. It raises the dollar cost of crude oil imports, which mechanically widens the trade deficit and hence the CAD (Section 6) - a real-side, price-and-volume mechanism, not an identity-driven one. Independently, the same energy-price shock raises the cost of fertiliser and LPG that the government has committed to subsidise, which widens the fiscal deficit (Section 7) - a policy-and-budget mechanism. Both channels share a root cause without either causing the other.

Evidence supporting this mechanism: Sections 6 and 7 quantify each channel independently, using separate data (crude import volumes and prices for the CAD channel; historical 2022 subsidy-overrun calibration for the fiscal channel). The fact that both estimates can be derived without reference to each other - and do not require assuming the identity - is itself evidence that the identity is not doing the causal work; the shared oil-price shock is.

Figure 1 below makes this distinction explicit.

Figure 1: Twin-Deficit Transmission Mechanism — External and Fiscal Channels



Source: Author's construction based on PPAC, RBI and Union Budget data; scenario framework per Section 5.

As Figure 1 shows, the identity (left branch) and the causal mechanism (right branch) are two separate statements. The Section 9 integrated analysis and Section 10 scenario tables that follow are built entirely on the causal, **common-shock mechanism** (right branch) - the identity is presented here only to pre-empt the error of citing it as proof, precisely because this is a common analytical mistake in twin-deficit discussions.

4.2 Why the Middle East Specifically Matters More Than a Generic Oil Shock

Three features of this conflict make it more dangerous than a routine, demand-driven oil-price rise. Chokepoint risk: the **Strait of Hormuz** carries roughly one-fifth of global oil supply, so even a partial, temporary disruption can cause a large price spike through risk premia alone, without any actual loss of physical supply. Compound shipping and energy shock: this situation combines an energy-price channel with a logistics channel (Red Sea disruption), each adding separate, additive costs to India's import bill - unlike a simple demand-driven price rise. Unique labour-market exposure: no prior major oil shock in India's recent history has coincided with such a large share of the workforce (~9 million) living in the conflict region, making the remittance channel (Section 6.3) a materially new amplifying factor.

5. Institutional and Stakeholder Analysis

The channels quantified in Section 4 do not operate through markets alone; each is administered by a specific institution with its own statutory mandate, policy levers and binding constraints. Understanding who can act, on what lever, and under what limitation is a precondition for the policy recommendations in Section 13. Table 5.1 maps the five institutions most exposed to this shock.

Institution	Primary Lever in This Shock	Binding Constraint	Interaction With the Other Deficit
Reserve Bank of India (RBI)	FX intervention (spot/forward) to smooth INR volatility; MPC rate decisions	2-6% CPI inflation-targeting mandate; reserves are finite and must also cover Week 3 capital-flight contingencies	A weaker Rupee (tolerated to preserve reserves) raises the landed cost of subsidised fertiliser and LPG, feeding back into the fiscal channel (Section 7)
Ministry of Finance (MoF)	Subsidy budgeting; excise duty settings; sovereign borrowing programme	FRBM-linked fiscal glide path (4.50% of GDP by FY26) monitored by rating agencies	Any consumer-price relief MoF funds via subsidy or excise cuts (Section 7) is fiscal absorption of a cost that would otherwise fall on the external account via demand destruction
Ministry of Petroleum & Natural Gas (MoPNG) / OMCs	Retail pricing pass-through decisions; SPR release; sourcing diversification	OMC balance-sheet capacity to absorb under-recovery (Section 8); SPR capacity is small relative to major global holders	Determines how much of the CAD-side import-bill shock (Section 6) is passed to consumers versus absorbed as fiscal/quasi-fiscal cost (Section 7-8)
Department of Fertilisers / Ministry of Chemicals & Fertilizers	Fertiliser subsidy disbursement and urea pricing administration	Disbursement timing lags price spikes, creating working-capital stress ahead of a	Directly administers the single largest fiscal-side channel

Institution	Primary Lever in This Shock	Binding Constraint	Interaction With the Other Deficit
		cropping season (Section 8)	quantified in this report (Section 7.2)
Ministry of External Affairs (MEA)	Gulf labour-market monitoring; worker welfare and evacuation planning	No conventional monetary or fiscal tool over remittance flows; depends on host-country conditions outside India's control	Monitors the CAD-side remittance channel (Section 6.3) that RBI and MoF cannot directly manage

No single institution can manage this shock alone: RBI's exchange-rate stance changes the fiscal cost of subsidies, MoF's subsidy and excise choices change the pressure on the currency, and MoPNG's pass-through decisions determine how the cost is split between the two deficits in the first place. This interdependence is the operational reason Section 13's recommendations are framed jointly rather than institution-by-institution.

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6. Oil Shock and the Current Account

6.1 Additional Crude Oil Import Bill and CAD Impact

Formula: Change in Import Bill (USD billion) = Change in Price (USD/bbl) × Quantity (billion barrels/year). Net crude oil import volume (FY24 reference) is 232.5 million tonnes; using a standard conversion of 7.33 barrels/tonne gives 1,704.3 million barrels/year (~1.70 billion barrels/year) - cross-checked against India's estimated net import rate of 4.7-4.9 million barrels/day (5.3-5.5 million barrels/day consumption less 0.6 million barrels/day domestic production), which is consistent within a reasonable margin.

This report uses the same unadjusted price-times-volume approach as the Week 1 report's headline figure, without applying a demand-elasticity offset. Week 1's Stage 2 sensitivity check shows this offset is small (under 0.5% of volume at a −0.05 short-run PED), so the methodological difference between the two reports does not materially change the conclusion - it is disclosed here for consistency rather than because it changes the result (see also Section 16.E).

Scenario	Change in Price (USD/bbl)	Q (billion bbl)	Change in Oil Import Bill (USD bn)	New CAD (USD bn)	New CAD (% GDP)	Change in from Base (pp)
Base	-	1.70	-	46	1.18%	-
A - Moderate	+10	1.70	17.0	63	1.62%	+0.44
B - Severe	+20	1.70	34.0	80	2.05%	+0.87

Scenario	Change in Price (USD/bbl)	Q (billion bbl)	Change in Oil Import Bill (USD bn)	New CAD (USD bn)	New CAD (% GDP)	Change in from Base (pp)
C - Extreme	+30	1.70	51.0	97	2.49%	+1.31

Scenario A pushes the CAD from a comfortable sub-1.5% zone into the 1.6% range - manageable, but reducing the safety buffer built up since FY24's 0.7% reading. **Scenario B** crosses **the 2% threshold that historically draws closer market and rating-agency scrutiny**, though it remains well below the ~4.8% peak seen during 2012-13. **Scenario C** approaches 2.5% of GDP, a level associated with higher external vulnerability. This calculation is consistent with the standard market rule of thumb of 0.3-0.5 percentage points of CAD/GDP per $10/barrel move, which corroborates the model.

Figure 2: Current Account Transmission Chain (Scenario A, +$10/bbl)



Source: Author's calculations based on PPAC import volumes and RBI CAD baseline; Scenario A values shown.

Figure 2 traces **Scenario A** end-to-end: a $10/barrel Brent shock raises the annual import bill by $17.0 billion, which widens the CAD by 0.44 percentage points of GDP, which in turn raises the foreign-currency financing requirement and puts pressure on external financing and the Rupee - the capital-flow and currency consequences of this pressure are developed fully in the Week 3 report.

6.2 Shipping and Logistics Overlay

Independent of the oil-price channel, Red Sea disruption raises freight costs on India-Europe trade (bilateral goods trade of ~USD 130 billion). Using an illustrative assumption that freight costs represent 3-5% of traded goods value, and that severe disruption periods have historically raised freight rates on affected routes by 50-100%: Additional freight cost ~ USD 130 billion × 4% (assumed freight share) × 75% (assumed mid-point rate increase) ~ USD 3.9 billion. This is materially smaller than the oil-price channel but not negligible, and it affects non-oil trade competitiveness - a distinct, additional drag compounding the oil-driven CAD pressure, most relevant in Scenarios B and C.

6.3 Remittance Channel Overlay

Gulf-origin remittances constitute roughly **USD 63 billion** of India's ~**USD 125 billion** annual remittance inflow. A 10% disruption to this corridor - from labour-market stress, hiring freezes, or wage stagnation among the ~9 million-strong Gulf-based Indian workforce - implies: Change in Remittances = **USD 63**

billion × 10% = USD 6.3 billion reduction; additional CAD stress = USD 6.3 billion / USD 3,900 billion ~ 0.16 percentage points of GDP. This channel is genuinely under-appreciated in generic oil-shock analyses: it is not captured in the price scenarios above and should be treated as an additive overlay, particularly relevant to **Scenario C**. The Ministry of External Affairs and Ministry of Finance should jointly monitor this exposure, since it is not a channel the RBI can manage through conventional monetary tools.

6.4 Forex Reserve Adequacy - Summary View

Import cover (Reserves ÷ [Annual Imports ÷ 12]) declines only modestly across scenarios: from **9.8 months** at baseline (**USD 655 billion** reserves against ~USD 800 billion annual imports) to **9.2 months** under **Scenario C**. This remains comfortably above **the conventional 3-month adequacy benchmark** and represents a materially stronger position than during the 2013 Taper Tantrum, when cover was closer to 7 months against a smaller import base. A full multi-metric reserve adequacy assessment - including the Guidotti-Greenspan ratio and IMF ARA framework, and the capital-flight scenarios that could draw reserves down faster than the import-cover measure alone implies - is the subject of the Week 3 report and is not repeated in full here.

7. Oil Shock and the Fiscal Position

7.1 Historical Anchor - The 2022 Precedent

Rather than estimating fertiliser and LPG subsidy sensitivity from first principles, this report anchors its estimate to the 2022 Russia-Ukraine energy crisis, an episode of comparable scale to a severe Middle East escalation. During that shock, India's fertiliser subsidy bill rose from a budgeted Rs 1,05,000 crore (FY23) to an actual outturn of approximately Rs 2,51,000 crore - an overrun of about Rs 1,46,000 crore, driven mainly by a near-doubling of global urea and fertiliser prices following natural-gas price spikes.

This report uses a more conservative estimate than the 2022 episode for three reasons: urea pricing and subsidy payment systems have since been updated; a larger share of India's LNG is now covered by long-term contracts, reducing spot-price exposure; and the 2022 shock was compounded by a Russia/Belarus potash supply disruption that is not a factor in a Middle East-centred scenario. Accordingly, this report estimates a Rs 15,000 crore fertiliser subsidy increase per $10/barrel of **sustained Brent** price rise - a defensible, conservative planning assumption rather than a direct extrapolation of the 2022 outturn.

Table 7.1a below makes the evidence hierarchy explicit for each subsidy channel: Observed FY25 baseline (Section 3, Table 3.1) -> 2022 historical precedent (Section 7.1, Table 11 H1/H3) -> conservative adjustment (author judgement, explained above) -> scenario-calibrated sensitivity (Table 14, S1-S2) -> resulting fiscal estimate (Table 7.3). Each step is traceable to a labelled source so the reasoning can be reproduced rather than taken on trust.

Channel	2022 Observed/Historical Value	Current FY25 Baseline	Adjustment/Assumption Applied	Modelled Scenario Range (A -> C)
Fertiliser subsidy	Rs 2,51,000 cr actual FY23 outturn vs Rs 1,05,000 cr BE (overrun ~Rs 1,46,000 cr) [Table 11, D11]	Rs 1,64,000 cr, FY25 BE [Table 11, D9]	Conservative discount vs. 2022 for post-2022 pricing/contract reforms (Section 7.1); Rs 15,000 cr assumed per $10/bbl [Table 14, S1]	Rs 15,000 - 42,000 cr [Table 5]
LPG/petroleum (OMC compensation)	Rs 22,000 cr one-time OMC compensation grant, 2022 [Table 11, D12]	No standing FY25 compensation line (ad hoc, event-driven)	One-time 2022 grant rescaled into a recurring per-$10/bbl sensitivity, Rs 18,000 cr/$10bbl [Table 14, S2]	Rs 18,000 - 52,000 cr [Table 5]

Two limitations of the 2022 anchor are worth stating directly. First, it is a single historical episode rather than a distribution of comparable shocks, so the resulting sensitivities (S1, S2) are a defensible planning calibration, not a statistically estimated elasticity. Second, the 2022 OMC compensation was a one-time payment rather than a recurring subsidy line; rescaling it into a per-$10/bbl sustained-scenario sensitivity is a modelling choice, not a direct historical read-across - this is why Scenario C's modelled LPG burden (Rs 52,000 crore) exceeds the single 2022 grant (Rs 22,000 crore). Both points are disclosed here so the calibration can be judged on its stated assumptions rather than mistaken for an observed value.

7.2 Fertiliser and LPG/Petroleum Subsidy Escalation

Scenario	Change in Price (USD/bbl)	Fertiliser Subsidy (Rs cr)	LPG/Petroleum Subsidy or OMC Compensation (Rs cr)	Total (Rs cr)
A	+10	15,000	18,000	33,000
B	+20	29,000	34,000	63,000
C	+30	42,000	52,000	94,000

The LPG/petroleum estimate is calibrated against the 2022 precedent of a **one-time Rs 22,000 crore** government grant to OMCs to cover under-recovery from holding domestic cylinder prices below import cost, scaled to this report's price scenarios.

7.3 Fiscal Deficit Impact

Formula: Change in **Fiscal Deficit** (% of GDP) = Total Additional Subsidy Burden (Rs crore) ÷ Nominal GDP (Rs crore) × 100, using FY25 GDP of Rs 32,640,000 crore.

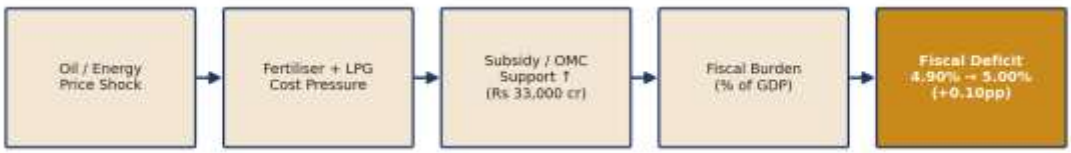
Scenario	Total Subsidy Burden (Rs cr)	Change in Fiscal Deficit (pp of GDP)	New Fiscal Deficit (% of GDP)
Base (FY25 BE)	-	-	4.90%
A (+$10/bbl)	33,000	0.10	5.00%
B (+$20/bbl)	63,000	0.19	5.09%

Scenario	Total Subsidy Burden (Rs cr)	Change in Fiscal Deficit (pp of GDP)	New Fiscal Deficit (% of GDP)
C (+$30/bbl)	94,000	0.29	5.19%

Sample calculation (**Scenario A**): Change in Fiscal Deficit = 33,000 ÷ 32,640,000 × 100 = 0.1011% ~ 0.10 percentage points; New Fiscal Deficit = **4.90%** + 0.10% = **5.00% of GDP**. The additional burden is real and significant in absolute rupee terms (**Rs 33,000-94,000 crore**) but modest as a share of GDP, given the scale of the denominator. The larger risk is the signalling effect on **fiscal credibility** relative to the FY25-FY26 glide path toward 4.5% of GDP, which rating agencies monitor closely. A slip of **10-29 basis points is manageable** through spending reprioritisation or non-tax revenue, provided the government avoids the substantially costlier step of cutting fuel excise duties (Section 7.4).

Reproducibility check: Scenario C's modelled fertiliser addition (Rs 42,000 crore, Table 5) is approximately 29% of the actual 2022 fertiliser subsidy overrun (Rs 1,46,000 crore, Table 7.1a) - i.e. 42,000 / 1,46,000 = 0.29 - which is the arithmetic expression of the conservative-discount rationale in Section 7.1, rather than an unsupported assertion of conservatism.

Figure 3: Fiscal Transmission Chain (Scenario A, +$10/bbl)



Source: Author's calculations calibrated to the 2022 fertiliser/LPG subsidy episode; Union Budget FY25 baseline.

7.4 The Excise Duty Risk (Excluded From the Baseline Scenarios)

One major fiscal risk is intentionally excluded from the scenarios above because it is a discretionary policy choice rather than an automatic cost: the temptation to cut fuel excise duties to lower pump prices. The May 2022 excise cut (Rs 8/litre on petrol, Rs 6/litre on diesel) is estimated to have cost the government approximately Rs 1,00,000 crore per year in lost tax revenue - **a single-year cost exceeding this report's entire Scenario C subsidy estimate**. If political pressure forces a repeat of this measure, the true fiscal cost would substantially exceed the figures modelled above; this risk is addressed directly in the policy trade-offs of Section 12.

8. Subsidy and Quasi-Fiscal Channels

Beyond the direct budgetary subsidy lines quantified in Section 7, two **quasi-fiscal** channels deserve explicit note because they do not always appear as an immediate budget-line increase but still represent a real economic cost borne by the state or state-owned entities. First, **OMC under-recovery**: when the government resists full retail pass-through of higher crude costs (Section 7.4), the cost does not disappear - it accumulates on OMC balance sheets as under-recovery, eventually requiring compensation (as in the 2022 Rs 22,000 crore grant) or weakening OMC creditworthiness. Second, fertiliser-sector working-capital stress: delayed subsidy disbursement during a price spike can strain fertiliser company working capital even before the budget line is formally revised, with knock-on effects for input availability ahead of a cropping season. Both channels reinforce why this report treats the fiscal impact (Section 7) as a floor on the true quasi-fiscal cost rather than a ceiling.

9. Integrated Twin-Deficit Analysis

Bringing Sections 6 and 7 together, Table 9.1 shows both deficits - and an author-constructed composite - the Author's Composite Twin-Deficit Stress Index (CAD % of GDP + Fiscal Deficit % of GDP), used in this report purely as a single-number summary of combined external-plus-fiscal pressure. This index is not a standard, internationally recognised macroeconomic metric; it is an analytical device introduced for this report, defined explicitly in Section 16.E, and should not be cited as an official RBI, IMF or Ministry of Finance measure.

Metric	Base	Scenario A	Scenario B	Scenario C
Brent crude shock	-	+$10/bbl	+$20/bbl	+$30/bbl
Additional oil import bill (USD bn)	-	17	34	51
Current Account Deficit (% GDP)	1.18%	1.62%	2.05%	2.49%
Additional subsidy burden (Rs cr)	-	33,000	63,000	94,000
Fiscal Deficit (% GDP)	4.90%	5.00%	5.09%	5.19%
Twin-Deficit Stress Index (composite, % GDP)	6.08%	6.62%	7.14%	7.68%
Import cover (months)	9.8	9.6	9.4	9.2
Illustrative INR pressure (per USD)	-	~85.3	~86.6	~87.9

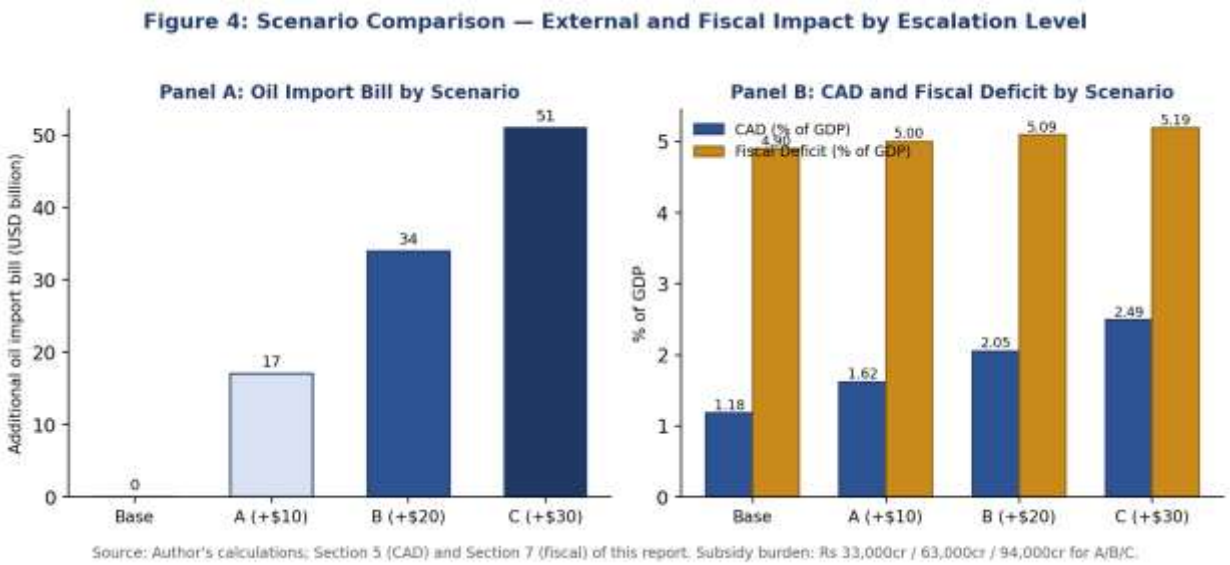


Figure 4 shows the two channels side by side: Panel A confirms the oil import bill scales linearly with the price shock (as expected, given the fixed-volume assumption in Section 6.1), while Panel B shows both the CAD and Fiscal Deficit ratios rising together across scenarios - the empirical signature of the **common-shock mechanism** established in Section 4, not of one deficit driving the other.

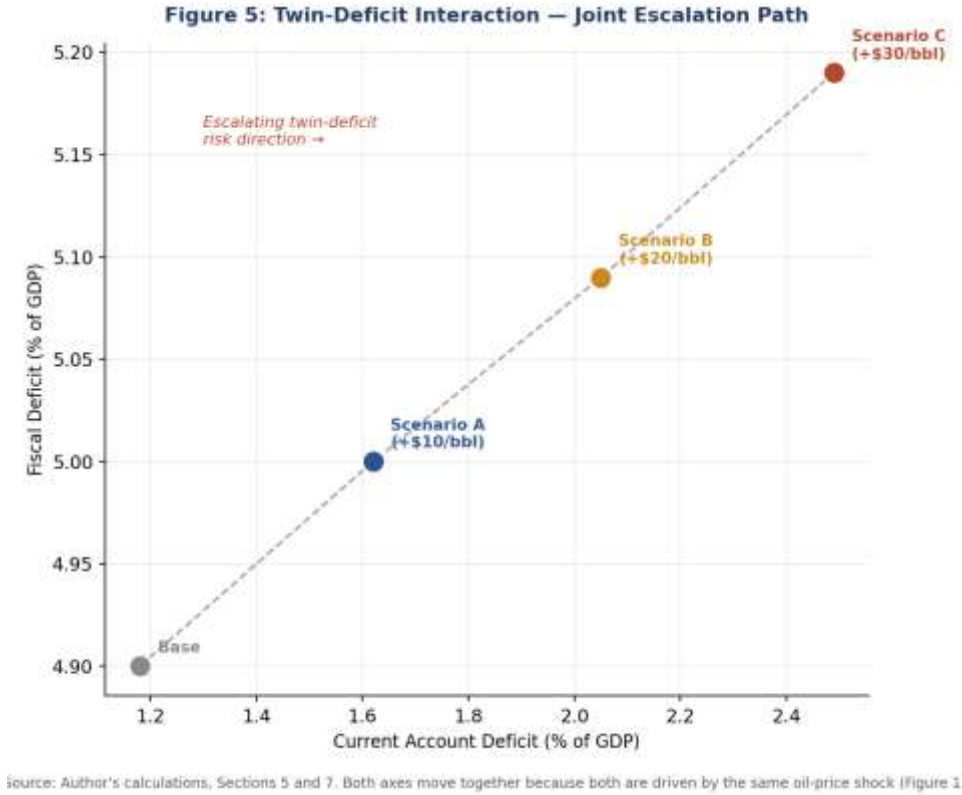


Figure 5 plots the same data as a joint escalation path. The near-linear co-movement of the CAD and Fiscal Deficit across scenarios is visually striking, and it is precisely this pattern that makes the twin-deficit identity (Section 4.1) tempting to over-interpret as causal. As established there, the correct reading is that

both series respond to the same underlying Brent price shock along the x- and y-dimensions independently - the chart shows correlation consistent with a common cause, not evidence of a causal link between the two axes themselves.

10. Scenario Analysis

The three scenarios - Moderate (**+$10/bbl**), Severe (**+$20/bbl**) and Extreme (**+$30/bbl**) - are constructed as a deliberate escalation ladder rather than a probability-weighted forecast, consistent with the Week 1 report's scenario framing. **Scenario A** represents a plausible, near-term escalation; **Scenario B** corresponds to a genuine **Strait of Hormuz** disruption risk; **Scenario C** represents a tail-risk, prolonged-conflict case.

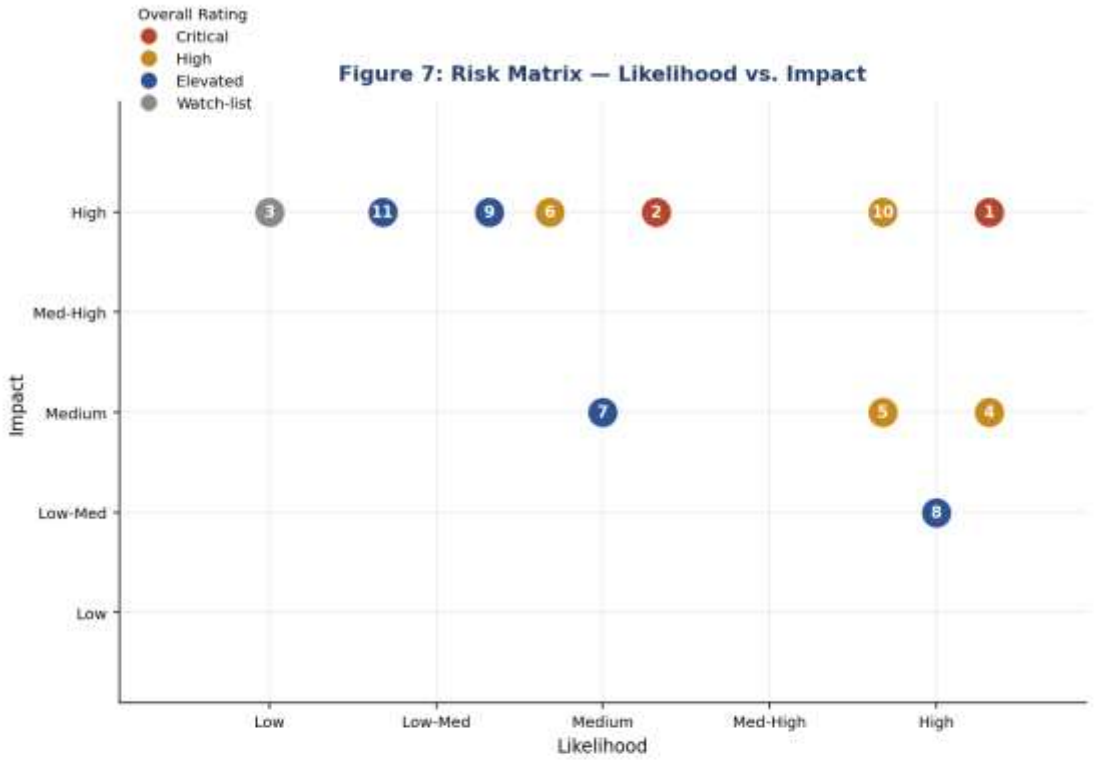
Across all three scenarios, two findings hold consistently: first, the fiscal deficit impact (**10-29 basis points**) is small relative to the CAD impact (**44-131 basis points**) as a share of GDP, because the subsidy channel, while large in absolute rupee terms, is a smaller fraction of India's Rs 326 lakh crore economy than the dollar-denominated import bill is of India's smaller (in rupee-equivalent terms per unit of GDP) trade account; and second, the **Twin-Deficit Stress Index** rises from **6.08% (Base)** to **7.68% (Scenario C)**, a level that - while not historically extreme in isolation - would combine with the currency and **capital-flow pressure**s analysed in the Week 3 report to materially raise the bar for policy credibility, the subject of Section 12.

11. Sensitivity and Risk Analysis

11.1 Risk Matrix

#	Risk	Likelihood	Impact	Rating	Primary Owner
1	Sustained Brent rise ($10-20/bbl)	High	High	Critical	RBI / MoF (joint)
2	Strait of Hormuz partial disruption	Medium	Very High	Critical	MoF / MoPNG / MEA
3	Full Hormuz closure (tail risk)	Low	Severe	Watch-list	Government of India
4	Fertiliser subsidy overrun	High	Medium	High	Ministry of Chemicals & Fertilizers
5	LPG price freeze / OMC under-recovery	High	Medium	High	MoPNG / MoF
6	INR depreciation beyond comfort band	Medium	High	High	RBI

#	Risk	Likelihood	Impact	Rating	Primary Owner
7	Gulf remittance disruption	Medium	Medium	Elevated	MEA / MoF
8	Red Sea shipping cost escalation	High	Low-Medium	Elevated	Ministry of Commerce / Shipping
9	Sovereign rating outlook pressure	Low-Medium	High	Elevated	MoF (DEA)
10	Excise duty cut political pressure	High	High	High	MoF / Cabinet
11	Capital flight / FPI debt outflow	Low-Medium	High	Elevated	RBI / SEBI



Source: Author's qualitative risk assessment, Section 10. Numbered points correspond to Table 10.1.

Figure 7 shows that the primary risks are not the catastrophic, low-probability tail events (a full Hormuz closure, Risk 3), but a cluster of **high-probability, moderate-to-high-impact risks** - sustained (not extreme) oil price increases, subsidy overruns, and political pressure to cut excise duties (Risks 1, 4, 5, 10). This has a direct implication for Section 12: policy preparation should prioritise resilience against these 'grinding, sustained' scenarios rather than concentrating all planning on the extreme tail, while still keeping contingency plans available for it.

11.2 Sensitivity to Key Assumptions

The Section 7 fiscal estimates are most sensitive to the fertiliser subsidy pass-through rate (Rs 15,000 crore per $10/barrel), which is calibrated conservatively against the 2022 episode. A doubling of this rate (reflecting a scenario where subsidy administration reforms since 2022 prove less effective than assumed) would raise **Scenario C**'s fiscal deficit impact **from 0.29pp to approximately 0.44pp of GDP** - still well within manageable range but eroding more of the FY26 glide-path buffer. The Section 6 CAD estimates are comparatively less sensitive to modelling choices, since the price-times-volume calculation rests on well-documented import volumes rather than a calibrated behavioural elasticity.

Table 11.2 formalises this stress test across a wider range of the pass-through assumption, holding the LPG/petroleum channel and the CAD estimate fixed, to show how much of the Scenario C fiscal and composite-index result depends on this single calibrated input.

Fertiliser Pass-Through Multiplier	Basis	Scenario C Fiscal Deficit Impact (pp of GDP)	New Fiscal Deficit (% of GDP)	Twin-Deficit Stress Index (Scenario C)
0.5x (downside)	Illustrative linear extension below the report's base calibration, for symmetry with the upside case	~0.15	~5.05%	~7.54%
1.0x (report base case)	Section 7.1 conservative calibration against the 2022 precedent	0.29	5.19%	7.68%
2.0x (reforms prove less effective than assumed)	Doubling of the fertiliser pass-through rate, Section 11.2	~0.44	~5.34%	~7.83%

Across a fourfold range in the pass-through assumption (0.5x-2.0x), Scenario C's Twin-Deficit Stress Index moves by less than 0.3 percentage points (7.54%-7.83%). This confirms the report's qualitative conclusion - both deficits widen materially under a severe shock - is robust to substantial uncertainty in the exact subsidy calibration, even though the precise rupee estimate is not.

11.3 CAD Sensitivity to Other External Variables

Section 6 isolates the oil-price channel. Table 11.3 places that channel alongside four other variables capable of moving the CAD by a comparable order of magnitude, so that the oil-price focus of this report can be justified quantitatively rather than assumed.

Variable	Shock Magnitude	Base Value (USD bn)	Impact (USD bn)	Change in CAD/GDP (pp)
Brent crude price	+$10/bbl	1.70 bn bbl/yr	+17.0	+0.44

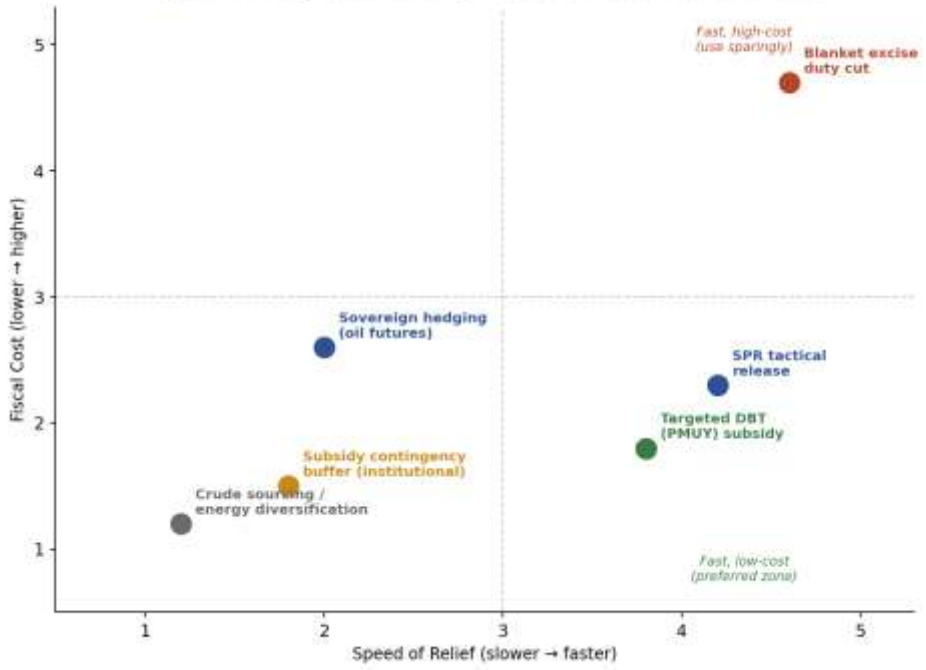
Variable	Shock Magnitude	Base Value (USD bn)	Impact (USD bn)	Change in CAD/GDP (pp)
Non-oil export growth	-2pp	~300 (non-oil exports)	-6.0	+0.15
Remittances (Gulf corridor)	-10%	~63	-6.3	+0.16
Gold imports	+10%	~45	+4.5	+0.12
Net FPI debt outflow (financing, not CAD)	-USD 10 bn	-	-10.0 (reserve draw)	n/a - BoP financing effect

The oil-price shock is roughly three times as impactful, per unit of shock, as the non-oil-export or remittance channels considered alone, which is the quantitative justification for this report's focus on the oil channel as primary. But the remittance and non-oil-export channels are not independent risks: both are plausibly correlated with the same Middle East escalation that drives the oil-price shock, so a full stress test should treat oil price, remittances and shipping costs (Section 6.2) as a correlated bundle rather than as separable, offsetting risks. The FPI-outflow row is included for completeness but is a balance-of-payments financing effect rather than a CAD driver in the accounting sense used elsewhere in this report; it is developed as a primary channel in the Week 3 report.

12. Policy Trade-offs

Every policy response to this shock involves a genuine trade-off between competing objectives. Figure 6 maps the main options by speed of relief against fiscal cost, which frames the four trade-offs discussed below.

Figure 6: Policy Trade-off Map — Speed of Relief vs. Fiscal Cost



Source: Author's assessment based on Section 7 (fiscal cost estimates) and Section 11 (policy trade-offs).

As Figure 6 shows, a blanket **excise duty** cut sits in the 'fast but high-cost' zone - it is the quickest lever available but the most expensive per Section 7.4. Targeted DBT subsidy delivery and SPR tactical release sit closer to the preferred fast-and-low-cost zone, while structural measures (diversification, a contingency buffer) are slower but carry little ongoing fiscal cost. This is the quantitative basis for the recommendations in Section 13.

Trade-off 1 - Growth versus Inflation Control (RBI)

Raising rates to fight energy-driven inflation helps price stability but slows growth; holding rates risks entrenched inflation drifting outside the RBI's 2-6% target band. This is the central Week 4 tension and is only briefly noted here.

Trade-off 2 - Consumer Protection versus Fiscal Discipline (MoF)

Holding retail LPG/fuel prices steady protects household income and headline inflation but transfers the cost to the fiscal deficit (Section 7); full pass-through protects the budget but produces a sharp, regressive inflation spike.

Trade-off 3 - Reserve Defence versus Reserve Preservation (RBI)

Aggressively defending the Rupee's level prevents depreciation but depletes reserves and can invite speculative testing; calibrated smoothing preserves reserves but allows some pass-through of a weaker Rupee into the subsidy bill (a feedback loop back into Section 7). Developed fully in the Week 3 report.

Trade-off 4 - Short-Term Relief versus Long-Term Credibility (MoF)

Excise cuts deliver immediate, visible relief but cost more than the subsidies modelled in Section 7 (Section 7.4) and erode the **fiscal glide path** that underpins rating-agency confidence; holding the line preserves credibility but is politically harder to sustain.

13. Policy Options, Recommendations & Roadmap

Each recommendation specifies the acting institution, mechanism, benefit, trade-off, timeline, and the quantitative indicator that would trigger escalation or reassessment.

R1 - RBI: Calibrated, non-defensive exchange-rate management

Smooth sharp INR movements without defending a fixed level; India's ~9-month **import cover** provides tactical room. Benefit: preserves market confidence without draining reserves. Trade-off: the Rupee still depreciates somewhat, raising import costs. Trigger: escalate to active defence only if import cover falls below **8 months** (cross-reference: Week 3 monitoring dashboard). Timing: Ongoing, applied continuously as INR volatility rises.

R2 - RBI/MoPNG: Expand local-currency (Rupee) trade settlement for energy imports

Extend the Rupee Vostro mechanism (used for Russian oil since 2022) to additional willing suppliers, reducing dollar demand. Benefit: lowers pressure on the Rupee. Trade-off: limited by counterpart willingness and trade-finance market depth. Trigger: review counterpart adoption quarterly; escalate diversification effort if USD/INR pressure (Section 6) exceeds **Rs. 86/USD** sustained. Timing: Begin counterpart outreach within one quarter; scale as adoption grows.

R3 - RBI: Inflation-vigilant monetary stance

Since fuel costs are ~13% of WPI, the MPC should use this report's scenarios to inform inflation forecasts and be ready to pause rate cuts under **Scenario B**-equivalent conditions. Trade-off: limits room to support growth. Trigger: pause cuts if projected 12-month CPI (Week 1, Section 9) exceeds **5.5%**. Benefit: keeps inflation expectations anchored within the RBI's 2-6% target band even as energy costs rise. Timing: Continuous, embedded in each MPC review cycle.

R4 - MoF: Preserve the fiscal glide path; resist blanket excise cuts

Favour narrowly targeted relief (continued PMUY DBT) over broad excise cuts, given the Section 7.4 cost differential. Benefit: preserves **fiscal credibility** at a fraction of the cost. Trade-off: relief is less visible to middle-class consumers outside targeted schemes. Trigger: escalate to partial excise modulation only if **Scenario B** materialises and sustains for a full quarter. Timing: Apply from the Budget cycle in which Scenario A-equivalent pressure is first observed.

R5 - MoF: Establish a dedicated Subsidy Contingency Buffer in the Union Budget

Pre-budget a contingency line sized to the Table 7.3 scenario range, avoiding disruptive mid-year reallocation as occurred in FY23. Benefit: smoother budget management, improved rating-agency predictability. Trade-off: requires showing a marginally higher headline deficit in 'normal' years. Timeline: FY26 Budget cycle onward. Trigger: Draw down the contingency line once the actual subsidy outturn exceeds the Budget Estimate by more than the Scenario A threshold (Rs 33,000 crore, Table 5).

R6 - MoPNG/MoF: Accelerate crude sourcing diversification and SPR utilisation rules

Continue expanding non-Gulf sourcing (Russia, US, Africa) and pre-define SPR release rules for price spikes. Benefit: reduces single-region concentration risk; smooths costs without permanent subsidy. Trade-off: India's SPR capacity remains small relative to major global holders. Timeline: 18+ months for capacity expansion; release rules can be set immediately. Trigger: Review diversification pace annually against realised Brent price volatility.

R7 - MoF/MEA: Formalise Gulf labour-market and remittance monitoring

Joint MoF-MEA tracking of Gulf employment conditions for Indian workers, with welfare/evacuation plans ready if **Scenario C** materialises. Benefit: protects both workers and a critical income source (Section 6.3). Timeline: immediate, permanent framework. Trade-off: requires sustained MEA staff time and data-sharing arrangements with host-country authorities that are largely outside India's control. Trigger: escalate to active welfare/evacuation planning if Scenario C materialises (Section 10).

R8 - RBI/MoF: Coordinated rating-agency and investor communication

Present a consistent, evidence-based narrative to credit rating agencies acknowledging the short-term twin-deficit deterioration (Section 9) while reaffirming the medium-term **fiscal glide path**. Benefit: reduces the risk of a disproportionate rating downgrade. Trigger: activate formally if the **Twin-Deficit Stress Index** (Section 9) exceeds **7.5%** (i.e., approaching the **Scenario C** composite level). Trade-off: requires disclosing the same twin-deficit deterioration analysed in this report to a market audience, carrying near-term signalling risk even though the medium-term narrative is reassuring. Timing: immediate, ahead of the next scheduled rating-agency review.

R9 - RBI: Contingency capital-flow facility (FCNR(B)-style)

Keep a targeted NRI-deposit swap scheme, on the design used in the 2013 **FCNR(B)** window, in a ready-to-activate state rather than building it from scratch under crisis conditions. Benefit: a proven, rapid-deployment financing buffer if **Scenario B** or C coincides with broader capital-flight pressure. Trade-off: requires regulatory and operational groundwork to be maintained on a standby basis with no current-year activation. Trigger: activate only if the Week 3 capital-flow monitoring dashboard signals sustained FPI debt outflow alongside a **Scenario B**-or-worse energy shock; this is a joint RBI-MoF contingency measure, not a routine tool, and is cross-referenced rather than re-derived here since the FCNR(B) precedent and mechanics are analysed in full in the Week 3 report. Timing: maintained on a standing standby basis; no current-year activation.

R10 - MoF: Feasibility study on sovereign oil-price hedging

Commission a feasibility study on hedging a defined portion of India's crude oil import exposure using futures or options, as several oil-importing sovereigns and national carriers already do. Benefit: converts part of an unpredictable subsidy/CAD exposure (Sections 6-7) into a budgeted, known cost, improving fiscal forecast reliability. Trade-off: requires specialised treasury expertise, and a poorly structured hedge can itself generate losses if prices fall rather than rise. Timeline: 6-12 months for a feasibility study before any pilot programme; this is a structural, not emergency, measure and sits alongside R6 in the Phase 3-4 roadmap below. Trigger: commission the study once Scenario B persists for two consecutive quarters, consistent with the R4 threshold.

13.1 Implementation Roadmap

Phase	Window	Focus	Lead Institutions
1 - Monitoring & Readiness	0-3 months	Track scenario indicators; prepare SPR release rules; brief investors; monitor Gulf employment	RBI, MoF, MEA
2 - Targeted Stabilisation	3-9 months	PMUY LPG discount continuity; calibrated currency management; inflation watch	RBI, MoPNG, MoF
3 - Fiscal Buffer Institutionalisation	6-18 months	Contingency budget line; explore sovereign hedging; expand Rupee trade settlement	MoF, RBI
4 - Structural De-risking	18+ months	SPR capacity expansion; energy diversification; remittance-corridor risk planning	MoPNG, MoF, MEA

The roadmap is front-loaded toward Phases 1-2, reflecting the Section 11 finding that high-probability, moderate-impact risks are the more urgent planning priority than rare worst-case events. Phase 4's structural measures are multi-year by nature and cannot substitute for near-term crisis management, but starting them immediately is the only way to reduce India's structural exposure to the next Middle East-driven energy shock.

14. Conclusion

The Middle East conflict reactivates a structural feature of India's economy: heavy reliance on imported oil, combined with a subsidy system that automatically absorbs part of any energy price spike. A moderate escalation (**+$10**/barrel) adds roughly **USD 17 billion** to the oil import bill and **Rs 33,000 crore** to the subsidy budget - individually manageable, but capable of pushing both the CAD and the **Fiscal Deficit** above their respective comfort zones simultaneously, which is the twin-deficit risk this report set out to quantify.

This co-movement is not evidence that either deficit causes the other - Section 4 establishes explicitly that the correct causal story runs through the shared **external shock**, not through the CAD-fiscal accounting identity. India's roughly 10 months of **import cover**, a large and still-growing economy, and direct 2022 experience managing a comparable shock together provide a genuine - though not unlimited - buffer. The remittance and shipping-cost channels (Section 6) add pressure that a simple oil-price test would miss, and the excise-duty temptation (Section 7.4) remains the single largest avoidable fiscal risk in this scenario set. If the RBI and Ministry of Finance respond with the targeted, coordinated measures set out in Section 13 - rather than reactive, broad-based ones - this shock is a manageable macroeconomic event rather than a self-inflicted crisis.

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16. Data, Evidence & Calculation Notes

This section consolidates the evidentiary basis for every material figure used in the report, separated by evidence type.

16.A Observed / Reported Data

ID	Variable	Value Used	Reference Period	Source
D1	Nominal GDP (FY25, BE)	Rs 326.4 lakh crore (~USD 3.90tn)	FY25 BE	Union Budget 2024-25
D2	Fiscal deficit target (FY25, BE)	4.90% of GDP	FY25 BE	Union Budget 2024-25
D3	Fiscal glide-path target (FY26)	4.50% of GDP	FY26 target	Budget Speech
D4	Current Account Deficit (FY24)	0.7% of GDP (~USD 23bn)	FY24	RBI
D5	Crude oil import dependency	87-88%	Latest annual	Ministry of Petroleum
D6	Net crude imports	232.5 million tonnes	FY24	PPAC
D7	Forex reserves	~USD 655 billion	FY25 reference	RBI
D8	USD/INR reference rate	~Rs. 84/USD	FY25 reference	Market reference
D9	Fertiliser subsidy (FY25, BE)	~Rs 1,64,000 crore	FY25 BE	Union Budget
D10	Total inward remittances	~USD 125 billion	Latest annual	World Bank / RBI
D11	2022 fertiliser subsidy outturn	~Rs 2,51,000 crore (vs Rs 1,05,000cr BE)	FY23	Union Budget disclosures
D12	2022 OMC compensation grant	Rs 22,000 crore	2022	Government of India announcement
D13	May 2022 excise cut revenue cost	~Rs 1,00,000 crore/year	FY23	Union Budget disclosures

16.B Historical Facts / Precedents

ID	Event	Period	Relevance
H1	2022 Russia-Ukraine energy shock - fertiliser/LPG subsidy overrun	FY23	Calibration anchor for Section 7 subsidy pass-through estimates.
H2	2013 Taper Tantrum - CAD peak and reserve cover	2012-13	Benchmark for how far current scenarios sit from a genuinely stressed external position.
H3	May 2022 fuel excise duty cut	2022	Evidence base for the Section 7.4 excise-cut cost risk.

16.C Calculated Values

ID	Metric	Result	Basis
C1	Annual crude import volume	~1.70 billion barrels/year	232.5 million tonnes × 7.33 bbl/tonne
C2	Oil import bill increase, Scenario A	USD 17.0 billion	$10 × 1.70bn barrels
C3	CAD, Scenario A	1.62% of GDP (+0.44pp)	(46 + 17) / 3,900 × 100
C4	Shipping/logistics overlay	~USD 3.9 billion	$130bn × 4% × 75%
C5	Remittance overlay (10% Gulf disruption)	~0.16pp of GDP	$6.3bn / $3,900bn × 100

ID	Metric	Result	Basis
C6	Fiscal deficit, Scenario A	5.00% of GDP (+0.10pp)	33,000 / 32,640,000 × 100
C7	Twin-Deficit Stress Index, Scenario A	6.62% (composite)	1.62% + 5.00%
C8	Import cover, Scenario C	9.2 months	655 / (851/12)

16.D Scenario / Modelling Inputs

ID	Variable	Scenario Value	Basis / Rationale
S1	Fertiliser subsidy sensitivity	Rs 15,000 crore per $10/bbl	Conservative fraction of the 2022 outturn, adjusted for post-2022 contract and pricing reforms.
S2	LPG/petroleum subsidy sensitivity	Rs 18,000 crore per $10/bbl	Scaled from the 2022 Rs 22,000 crore OMC compensation precedent.
S3	Freight-cost share of traded goods value	3-5% (4% used)	Illustrative assumption; no India-specific published elasticity.
S4	Severe-disruption freight rate increase	50-100% (75% mid-point used)	Historical range on comparable disrupted routes.
S5	FX sensitivity (illustrative INR pressure)	~Rs. 85.3 / Rs. 86.6 / Rs. 87.9 across A/B/C	Illustrative, consistent with Week 1's β-based sensitivity framework; not independently re-derived here.

16.E Methodological Notes and Limitations

- The Author's Composite Twin-Deficit Stress Index (CAD % of GDP + Fiscal Deficit % of GDP), used in Section 9, is an author-constructed analytical composite created for this report and used solely for comparative scenario analysis. It is not a standard IMF, RBI or Ministry of Finance metric and should not be cited as one; it is used solely as a single-number summary of combined external-plus-fiscal pressure across scenarios.
- Section 4.1 explicitly distinguishes the CAD/fiscal accounting identity from the causal mechanism used in this report's analysis. All scenario calculations in Sections 6, 7, 9 and 10 are derived independently for each channel (import bill for CAD; subsidy calibration for fiscal deficit) and do not rely on the identity to generate either figure.
- This report uses the same unadjusted price-times-volume approach to the oil import bill as the Week 1 report's headline figure, rather than Week 1's supplementary elasticity-adjusted calculation. Week 1's own sensitivity check shows this difference is immaterial (under 0.5% of volume), so the two reports' figures are consistent to within rounding.
- The fiscal estimates (Section 7) are calibrated against a single historical precedent (2022); while adjusted conservatively for known structural changes since then, they should be read as defensible planning estimates rather than econometrically derived forecasts.
- This report does not independently re-derive the exchange-rate sensitivity used for the 'illustrative INR pressure' row in Section 9; it carries forward the β-based framework introduced in the Week 1 report for internal consistency, and the currency and reserve-adequacy analysis is developed in full in the Week 3 report.

Appendix: Summary of Key Formulas

This appendix consolidates the formulas used across Sections 6, 7, 9 and 11 in one place for quick reference; each formula is derived and applied in full, with worked examples, in its source section.

Calculation	Formula	Source Section
Oil import bill impact	Change in Bill (USD bn) = Change in Price (USD/bbl) x Quantity (bn bbl/yr)	6.1
CAD (% of GDP)	(Base CAD + Change in Bill) / Nominal GDP x 100	6.1
Shipping/logistics overlay	Bilateral trade value x Freight share x Rate increase	6.2
Remittance overlay	Gulf remittances x Disruption % / Nominal GDP x 100	6.3
Import cover	Reserves / (Annual Imports / 12)	6.4
Fiscal deficit (% of GDP) impact	Additional Subsidy Burden (Rs cr) / Nominal GDP (Rs cr) x 100	7.3
Twin-Deficit Stress Index (composite, author-constructed)	CAD (% GDP) + Fiscal Deficit (% GDP)	9 / 16.E
CAD sensitivity to non-oil variables	Base Value x Shock Magnitude / Nominal GDP x 100	11.3